

Proprioception

The body has its own sense of where it is and how it is moving in space. This process happens almost unconsciously, making it, in essence, the body's sixth sense.

Body position sense

Inside muscles, tendons, and joints are movement receptors called proprioceptors. Every time we move, these receptors measure changes in length, tension, and pressure that relate to that movement and send impulses to the brain. The information is processed and a decision is made to stop moving or change position. Messages are then relayed back to the muscles to carry out the decision. All of this happens without us having to think about it.

Types of proprioception

Most of the information our brain receives about body position is processed unconsciously, such as how we are constantly adjusting the position of our body to maintain balance. However, proprioceptive information can become conscious if it requires us to make a decision—for example, refining muscle movement to make a voluntary, skilled movement.

Proprioception pathways

Conscious proprioception signals travel up the brain stem to the thalamus and end at the parietal lobe, which is part of the cerebral cortex. The unconscious pathway loops back to the cerebellum, which controls movement.

